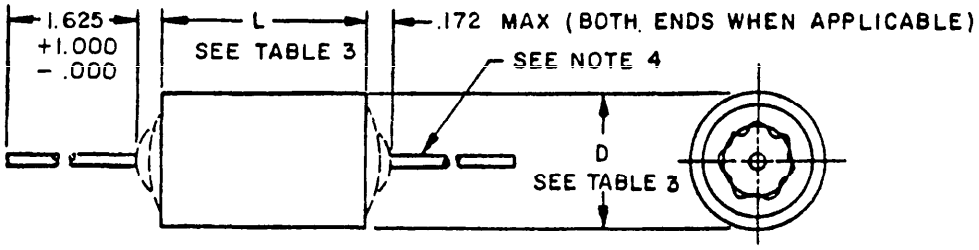
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, (HERMETICALLY SEALED IN METAL CASES), STYLES CQ08, CQ09, CQ12, AND CQ13

This specification is approved for use by all Departments and Agencies of the Department of Defense.

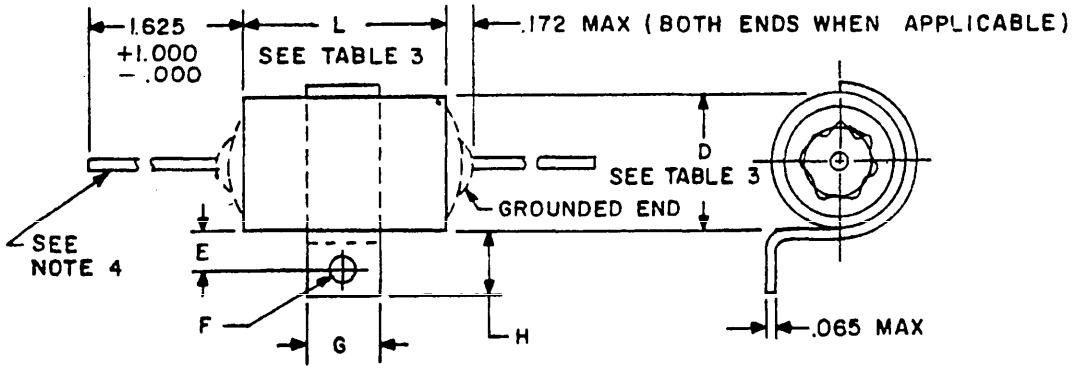
The complete requirements for acquiring the capacitors described herein shall consist of this document and the latest issue of MIL-C-19978.

(D) CHARACTERISTICS K AND M ARE INACTIVE FOR NEW DESIGN AFTER 23 OCTOBER 1973. FOR REPLACEMENT, USE COMPARABLE CQR STYLES OF MIL-C-19978/9, /10, /11, /13, /14, AND /15.



Style CQ08 (uninsulated case)

Style CQ09 (insulated case)



(D)

Case dimension D	H ±.062	G ±.062	E ±.031	F ±.005
Less than .500	.312	.250	.188	.144
.500 and up	.438	.500	.250	.156

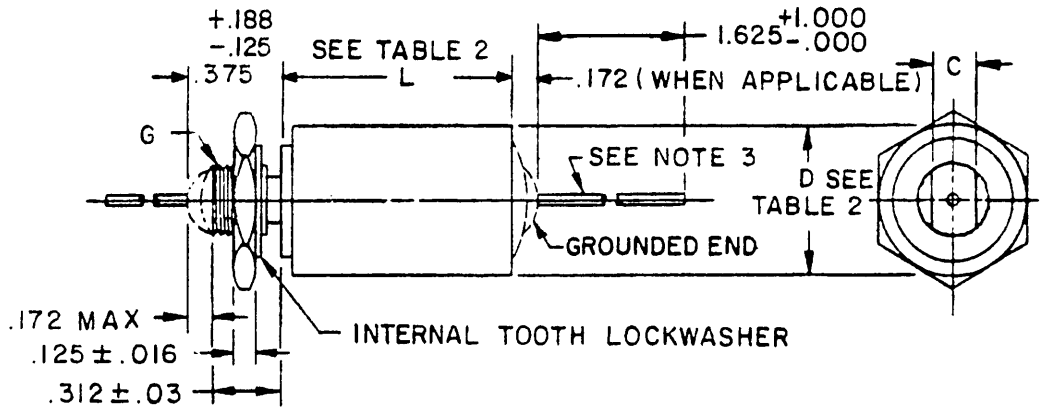
Style CQ12 (uninsulated case)

FIGURE 1. Styles CQ08, CQ09, CQ12 and CQ13 capacitors.

(D) denotes changes

FSC 5910

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INCHES	MM	INCHES	MM
.001	0.03	.188	4.78
.004	0.10	.195	4.95
.005	0.13	.235	5.97
.016	0.41	.250	6.35
.020	0.51	.312	7.92
.025	0.64	.375	9.53
.031	0.79	.400	10.16
.032	0.81	.437	11.10
.062	1.57	.438	11.13
.0625	1.59	.500	12.70
.125	3.18	.562	14.27
.144	3.66	.670	17.02
.156	3.96	.750	19.05
.172	4.37	1.000	25.40
.175	4.45	1.562	39.67
		1.625	41.28

Case dimension D	C ±.005	G
.562 or less	.250	5/16-24 UNF -2A
.670	.375	7/16-28 UNEF-2A
.750 and up	.437	1/2-28 UNEF-2A

Style CQ13 (uninsulated case)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Basic case dimensions for the insulated styles shall be the same as for the uninsulated styles.
4. Leads shall be of tinned solid wire, .020 (No. 24 AWG) for cases .175 and .195 in diameter; .025 (No. 22 AWG) for cases .235 through .312 diameter; and .032 (No. 20 AWG) for cases .400 diameter and above except that style CQ13 capacitors shall be furnished with .032 (No. 20 AWG) lead wire. Tolerance on all lead wire diameters shall be +.004, -.001.
(D) 5. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting such as tangential retainer for style CQ08, and a wrap-around hand for style CQ09. The supporting device will not be supplied with the capacitor. Style CQ13 shall be supplied with mounting nuts and lockwashers.
(D) 6. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Styles CQ08, CQ09, CQ12 and CQ13 capacitors - Continued.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table III.
Case material - Nonmagnetic (screwneck and end seal may be of magnetic material).
Characteristic - See table I.

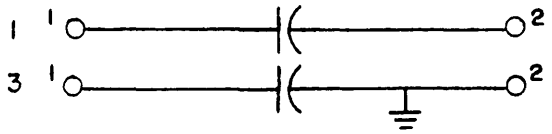
TABLE I. Characteristic.

Dielectric material	Characteristic symbol	Rated temperature	Dielectric absorption
(D) Paper and polyethelene terephthalate	K 1/	-65° to +125°C	---
Polyethelene terephthalate	M 1/	-65° to +85°C	0.60
Polystyrene	P	-65° to +65°C	0.10
Polytetrafluoroethylene	T	-65° to +170°C	0.10

1/ Characteristics K and M are inactive for new equipment design.

Terminals - Axial-wire lead (see figure 1).

Circuit diagram –



Rated voltage - See table III.
Rated temperature - See table I.
Voltage derating (characteristic K only) - See figure 2.
Capacitance (cap) (nom) - See table III.
Capacitance tolerance - See table III.
Dissipation factor (DF) (max) - Characteristic K - 1.0 percent.
Characteristic M - 0.6 percent.
Characteristics P and T - 0.1 percent.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Dielectric withstanding voltage (DWV):
Sleeving - In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Sleeving - In accordance with MIL-C-19978.
Terminal to terminal - See table II.
Terminal to case - In accordance with MIL-C-19978 except that characteristic K shall be greater than 10,000 megohms.

Vibration high frequency - Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

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TABLE II. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/
Characteristic M	At 25°C
0 to 0.68 microfarad	100,000 megohms
Greater than 0.68 microfarad	68,000 megohm-microfarads 1/
	At 85°C
0 to 0.2 microfarad	25,000 megohms
Greater than 0.2 microfarad	5,000 megohm-microfarads 1/
Characteristic P	At 25°C
0 to 1.0 microfarad	100,000 megohms
Greater than 1.0 microfarad	100,000 megohm-microfarads 1/
	At 65°C
0 to 1.0 microfarad	40,000 megohms
Greater than 1.0 microfarad	40,000 megohm-microfarads 1/
Characteristic T	At 25°C
0 to 1.0 microfarad	100,000 megohms
Greater than 1.0 microfarad	100,000 megohm-microfarads 1/
	At 170°C
0 to 0.2 microfarad	40,000 megohms
Greater than 0.02 microfarad	750 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Immersion:
  DWV:
    Sleeving - In accordance with MIL-C-19978.
  IR:
    Sleeving - In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

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Moisture resistance:
DWV:
Sleeving - In accordance with MIL-C-19978.
IR:
Sleeving - In accordance with MIL-C-19978.

Dielectric absorption (not applicable to values of 0.01 µF or less): See table I.

Stability at low and high temperatures:
Low temperature:
Test temperature - -65°C +0°, -5°C.

Capacitance change (max) - -10 percent for characteristic K.
-7 percent for characteristic M.
+5 percent for characteristic T.

High temperature:
Test temperature - +125°C +5°, -0°C for characteristic K.
+85°C +5°, -0°C for characteristic M.
+170°C +5°, -0°C for characteristic T.

Capacitance change (max) - +10 percent for characteristic K.
+5 percent for characteristic M.
-5 percent for characteristic T.

Temperature coefficient (characteristic P only): In accordance with MIL-C-19978.

Life:

Capacitance change (max): - ±5 percent of initial measured value for characteristics K, M, and T.
±6 percent of initial measured value for characteristic P with capacitance 0.05 µF and less.
±3 percent of initial measured value for characteristic P with capacitance greater than 0.05 µF.

Resistance to soldering heat: In accordance with MIL-C-19978.

TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors.

CHARACTERISTIC K				Case dimensions Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KC392-3	200	.0039	G, J, K	.750	.235	.688	.235 2/
CQ--A-KC472-3	200	.0047	G, J, K				
CQ--A-KC562-3	200	.0056	G, J, K				
CQ--A-KC682-3	200	.0068	G, J, K				
CQ--A-KC183-3	200	.018	G, J, K	.875	.312	.812	.312 2/
CQ--A-KC223-3	200	.022	G, J, K				
CQ--A-KC273-3	200	.027	G, J, K				
CQ--A-KC333-3	200	.033	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC K				Case dimensions Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KC393-3	200	.039	G, J, K	.875	.400	.812	.400
CQ--A-KC473-3	200	.047	G, J, K				
CQ--A-KC563-3	200	.056	G, J, K				
CQ--A-KC683-3	200	.068	G, J, K				
CQ--A-KC823-3	200	.082	G, J, K	1.125	.400	1.062	.400
CQ--A-KC104-3	200	.10	G, J, K				
CQ--A-KC124-3	200	.12	G, J, K	1.375	.400	1.312	.400
CQ--A-KC154-3	200	.15	G, J, K				
CQ--A-KC184-3	200	.18	G, J, K	1.125	.562	1.062	.562
CQ--A-KC224-3	200	.22	G, J, K				
CQ--A-KC274-3	200	.27	G, J, K	1.375	.562	1.312	.562
CQ--A-KC334-3	200	.33	G, J, K				
CQ--A-KC394-3	200	.39	G, J, K	1.625	.562	1.562	.562
CQ--A-KC474-3	200	.47	G, J, K				
CQ--A-KC564-3	200	.56	G, J, K	1.625	.670	1.562	.670
CQ--A-KC684-3	200	.68	G, J, K				
CQ--A-KC824-3	200	.82	G, J, K	2.125	.750	2.062	.750
CQ--A-KC105-3	200	1.000	G, J, K				
CQ--A-KE272-3	400	.0027	G, J, K	.750	.235	.688	.235 2/
CQ--A-KE332-3	400	.0033	G, J, K				
CQ--A-KE123-3	400	.012	G, J, K	.875	.312	.812	.312 2/
CQ--A-KE153-3	400	.015	G, J, K				
CQ--A-KE273-3	400	.027	G, J, K	.875	.400	.812	.400
CQ--A-KE333-3	400	.033	G, J, K				
CQ--A-KE393-3	400	.039	G, J, K	1.125	.400	1.062	.400
CQ--A-KE473-3	400	.047	G, J, K				
CQ--A-KE563-3	400	.056	G, J, K	1.375	.400	1.312	.400
CQ--A-KE683-3	400	.068	G, J, K				
CQ--A-KE823-3	400	.082	G, J, K	1.125	.562	1.062	.562
CQ--A-KE104-3	400	.10	G, J, K				
CQ--A-KE124-3	400	.12	G, J, K	1.375	.562	1.312	.562
CQ--A-KE154-3	400	.15	G, J, K				
CQ--A-KE184-3	400	.18	G, J, K	1.625	.562	1.562	.562
CQ--A-KE224-3	400	.22	G, J, K				
CQ--A-KE274-3	400	.27	G, J, K	1.625	.670	1.562	.670
CQ--A-KE334-3	400	.33	G, J, K				
CQ--A-KE394-3	400	.39	G, J, K	2.125	.750	2.062	.750
CQ--A-KE474-3	400	.47	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC K				Case dimensions Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KF102-3	600	.0010	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF122-3	600	.0012	G, J, K				
CQ--A-KF152-3	600	.0015	G, J, K				
CQ--A-KF182-3	600	.0018	G, J, K				
CQ--A-KF222-3	600	.0022	G, J, K				
CQ--A-KF272-3	600	.0027	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF332-3	600	.0033	G, J, K				
CQ--A-KF392-3	600	.0039	G, J, K				
CQ--A-KF472-3	600	.0047	G, J, K				
CQ--A-KF562-3	600	.0056	G, J, K				
CQ--A-KF682-3	600	.0068	G, J, K				
CQ--A-KF822-3	600	.0082	G, J, K				
CQ--A-KF103-3	600	.010	G, J, K				
CQ--A-KF123-3	600	.012	G, J, K	.875	.400	.812	.400
CQ--A-KF153-3	600	.015	G, J, K				
CQ--A-KF183-3	600	.018	G, J, K				
CQ--A-KF223-3	600	.022	G, J, K				
CQ--A-KF273-3	600	.027	G, J, K	1.125	.400	1.062	.400
CQ--A-KF333-3	600	.033	G, J, K				
CQ--A-KF393-3	600	.039	G, J, K	1.375	.400	1.312	.400
CQ--A-KF473-3	600	.047	G, J, K				
CQ--A-KF563-3	600	.056	G, J, K	1.125	.562	1.062	.562
CQ--A-KF683-3	600	.068	G, J, K				
CQ--A-KF823-3	600	.082	G, J, K	1.375	.562	1.312	.562
CQ--A-KF104-3	600	.10	G, J, K				
CQ--A-KF124-3	600	.12	G, J, K	1.625	.562	1.562	.562
CQ--A-KF154-3	600	.15	G, J, K				
CQ--A-KF184-3	600	.18	G, J, K	1.625	.670	1.562	.670
CQ--A-KF224-3	600	.22	G, J, K				
CQ--A-KF274-3	600	.27	G, J, K	2.125	.750	2.062	.750
CQ--A-KF334-3	600	.33	G, J, K				
CQ--A-KF394-3	600	.39	G, J, K	2.375	.750	2.312	.750
CQ--A-KF474-3	600	.47	G, J, K				
CQ--A-KG102-3	1000	.0010	G, J, K	.875	.400	.812	.400 5/
CQ--A-KG122-3	1000	.0012	G, J, K				
CQ--A-KG152-3	1000	.0015	G, J, K				
CQ--A-KG182-3	1000	.0018	G, J, K				
CQ--A-KG222-3	1000	.0022	G, J, K				
CQ--A-KG272-3	1000	.0027	G, J, K				
CQ--A-KG332-3	1000	.0033	G, J, K				
CQ--A-KG392-3	1000	.0039	G, J, K				
CQ--A-KG472-3	1000	.0047	G, J, K				
CQ--A-KG562-3	1000	.0056	G, J, K				
CQ--A-KG682-3	1000	.0068	G, J, K				
CQ--A-KG822-3	1000	.0082	G, J, K				
CQ--A-KG103-3	1000	.010	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC K - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KG123-3	1000	.012	G, J, K	1.125	.400	1.062	.400 5/
CQ--A-KG153-3	1000	.015	G, J, K				
CQ--A-KG183-3	1000	.018	G, J, K	1.375	.400	1.312	.400 5/
CQ--A-KG223-3	1000	.022	G, J, K				
CQ--A-KG273-3	1000	.027	G, J, K	1.125	.562	1.062	.562 5/
CQ--A-KG333-3	1000	.033	G, J, K				
CQ--A-KG393-3	1000	.039	G, J, K	1.375	.562	1.312	.562 5/
CQ--A-KG473-3	1000	.047	G, J, K				
CQ--A-KG563-3	1000	.056	G, J, K	1.625	.562	1.562	.562 5/
CQ--A-KG683-3	1000	.068	G, J, K				
CQ--A-KG823-3	1000	.082	G, J, K	1.625	.670	1.562	.670 5/
CQ--A-KG104-3	1000	.10	G, J, K				
CQ--A-KG124-3	1000	.12	G, J, K	1.875	.670	1.812	.670 5/
CQ--A-KG154-3	1000	.15	G, J, K				
CQ--A-KG184-3	1000	.18	G, J, K	2.125	.750	2.062	.750
CQ--A-KG224-3	1000	.22	G, J, K				
CQ--A-KG274-3	1000	.27	G, J, K	2.125	1.0	2.062	1.000
CQ--A-KG334-3	1000	.33	G, J, K				
CQ--A-KG394-3	1000	.39	G, J, K	2.375	1.0	2.312	1.000
CQ--A-KG474-3	1000	.47	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC M				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MZ155-3	30	1.5	G, J, K	1.625	.562	1.531	.562
CQ--A-MZ225-3	30	2.2	G, J, K	1.875	.670	1.781	.670
CQ--A-MZ335-3	30	3.3	G, J, K	1.875	.750	1.781	.750
CQ--A-MZ475-3	30	4.7	G, J, K	2.625	.750	2.531	.750
CQ--A-MZ685-3	30	6.8	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MZ106-3	30	10.0	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MA102-3	50	.0010	G, J, K				
CQ--A-MA122-3	50	.0012	G, J, K				
CQ--A-MA152-3	50	.0015	G, J, K				
CQ--A-MA182-3	50	.0018	G, J, K				
CQ--A-MA222-3	50	.0022	G, J, K				
CQ--A-MA272-3	50	.0027	G, J, K				
CQ--A-MA332-3	50	.0033	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA392-3	50	.0039	G, J, K				
CQ--A-MA472-3	50	.0047	G, J, K				
CQ--A-MA562-3	50	.0056	G, J, K				
CQ--A-MA682-3	50	.0068	G, J, K				
CQ--A-MA822-3	50	.0082	G, J, K				
CQ--A-MA103-3	50	.010	G, J, K				
CQ--A-MA123-3	50	.012	G, J, K				
CQ--A-MA153-3	50	.015	G, J, K	.500	.235	.406	.235 2/
CQ--A-MA183-3	50	.018	G, J, K				
CQ--A-MA223-3	50	.022	G, J, K				
CQ--A-MA273-3	50	.027	G, J, K				
CQ--A-MA333-3	50	.033	G, J, K	.500	.312	.406	.312 2/
CQ--A-MA393-3	50	.039	G, J, K				
CQ--A-MA473-3	50	.047	G, J, K				
CQ--A-MA563-3	50	.056	G, J, K	.562	.312	.468	.312 2/
CQ--A-MA683-3	50	.068	G, J, K				
CQ--A-MA124-3	50	.12	G, J, K	.875	.312	.781	.312 2/
CQ--A-MA154-3	50	.15	G, J, K				
CQ--A-MA184-3	50	.18	G, J, K	.875	.400	.781	.400
CQ--A-MA224-3	50	.22	G, J, K				
CQ--A-MA274-3	50	.27	G, J, K	1.000	.400	.906	.400
CQ--A-MA334-3	50	.33	G, J, K				
CQ--A-MA394-3	50	.39	G, J, K	1.125	.400	1.031	.400
CQ--A-MA474-3	50	.47	G, J, K				
CQ--A-MA564-3	50	.56	G, J, K	1.375	.400	1.281	.400
CQ--A-MA684-3	50	.68	G, J, K				
CQ--A-MA824-3	50	.82	G, J, K	1.125	.562	1.031	.562
CQ--A-MA105-3	50	1.0	G, J, K				
CQ--A-MA125-3	50	1.2	G, J, K	1.375	.562	1.281	.562
CQ--A-MB125-3	100	1.2	G, J, K				
CQ--A-MB155-3	100	1.5	G, J, K	1.625	.670	1.531	.670
CQ--A-MB185-3	100	1.8	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC M - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MB225-3	100	2.2	G, J, K	1.875	.750	1.781	.750
CQ--A-MB275-3	100	2.7	G, J, K	2.125	.750	2.031	.750
CQ--A-MB335-3	100	3.3	G, J, K	2.375	.750	2.281	.750
CQ--A-MB395-3	100	3.9	G, J, K	1.875	1.000	1.781	1.000
CQ--A-MB475-3	100	4.7	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MB565-3	100	5.6	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MB685-3	100	6.8	G, J, K	2.625	1.000	2.531	1.000
CQ--A-MC102-3	200	.0010	G, J, K				
CQ--A-MC122-3	200	.0012	G, J, K				
CQ--A-MC152-3	200	.0015	G, J, K				
CQ--A-MC182-3	200	.0018	G, J, K				
CQ--A-MC222-3	200	.0022	G, J, K				
CQ--A-MC272-3	200	.0027	G, J, K				
CQ--A-MC332-3	200	.0033	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC392-3	200	.0039	G, J, K				
CQ--A-MC472-3	200	.0047	G, J, K				
CQ--A-MC562-3	200	.0056	G, J, K				
CQ--A-MC682-3	200	.0068	G, J, K				
CQ--A-MC822-3	200	.0082	G, J, K				
CQ--A-MC103-3	200	.010	G, J, K				
CQ--A-MC123-3	200	.012	G, J, K	.719	.195	.625	.195 2/
CQ--A-MC153-3	200	.015	G, J, K				
CQ--A-MC183-3	200	.018	G, J, K				
CQ--A-MC223-3	200	.022	G, J, K	.719	.235	.625	.235 2/
CQ--A-MC273-3	200	.027	G, J, K				
CQ--A-MC333-3	200	.033	G, J, K				
CQ--A-MC393-3	200	.039	G, J, K	.719	.312	.625	.312 2/
CQ--A-MC473-3	200	.047	G, J, K				
CQ--A-MC563-3	200	.056	G, J, K				
CQ--A-MC683-3	200	.068	G, J, K	.844	.312	.750	.312 2/
CQ--A-MC823-3	200	.082	G, J, K				
CQ--A-MC104-3	200	.10	G, J, K				
CQ--A-MC124-3	200	.12	G, J, K	.812	.400	.718	.400
CQ--A-MC154-3	200	.15	G, J, K				
CQ--A-MC184-3	200	.18	G, J, K	1.062	.400	.969	.400
CQ--A-MC224-3	200	.22	G, J, K				
CQ--A-MC274-3	200	.27	G, J, K	1.250	.400	1.156	.400
CQ--A-MC334-3	200	.33	G, J, K				
CQ--A-MC394-3	200	.39	G, J, K	1.188	.500	1.094	.500
CQ--A-MC474-3	200	.47	G, J, K				
CQ--A-MC564-3	200	.56	G, J, K	1.375	.562	1.281	.562
CQ--A-MC684-3	200	.68	G, J, K				
CQ--A-MC824-3	200	.82	G, J, K	1.875	.562	1.781	.562
CQ--A-MC105-3	200	1.0	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC M - Continued				Case dimensions Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-ME102-3	400	.0010	G, J, K				
CQ--A-ME122-3	400	.0012	G, J, K				
CQ--A-ME152-3	400	.0015	G, J, K				
CQ--A-ME182-3	400	.0018	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME222-3	400	.0022	G, J, K				
CQ--A-ME272-3	400	.0027	G, J, K				
CQ--A-ME332-3	400	.0033	G, J, K				
CQ--A-ME392-3	400	.0039	G, J, K				
CQ--A-ME472-3	400	.0047	G, J, K				
CQ--A-ME562-3	400	.0056	G, J, K				
CQ--A-ME682-3	400	.0068	G, J, K				
CQ--A-ME822-3	400	.0082	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME103-3	400	.010	G, J, K				
CQ--A-ME123-3	400	.012	G, J, K				
CQ--A-ME153-3	400	.015	G, J, K				
CQ--A-ME183-3	400	.018	G, J, K	1.062	.312	.969	.312 2/
CQ--A-ME223-3	400	.022	G, J, K				
CQ--A-ME273-3	400	.027	G, J, K	.938	.400	.844	.400
CQ--A-ME333-3	400	.033	G, J, K				
CQ--A-ME393-3	400	.039	G, J, K	1.062	.400	.969	.400
CQ--A-ME473-3	400	.047	G, J, K				
CQ--A-ME563-3	400	.056	G, J, K	1.250	.400	1.156	.400
CQ--A-ME683-3	400	.068	G, J, K				
CQ--A-ME823-3	400	.082	G, J, K	1.562	.400	1.469	.400
CQ--A-ME104-3	400	.10	G, J, K				
CQ--A-ME124-3	400	.12	G, J, K	1.281	.562	1.188	.562
CQ--A-ME154-3	400	.15	G, J, K				
CQ--A-ME184-3	400	.18	G, J, K	1.531	.562	1.438	.562
CQ--A-ME224-3	400	.22	G, J, K				
CQ--A-ME274-3	400	.27	G, J, K	1.906	.562	1.812	.562
CQ--A-ME334-3	400	.33	G, J, K				
CQ--A-ME394-3	400	.39	G, J, K	1.906	.670	1.812	.670
CQ--A-ME474-3	400	.47	G, J, K				
CQ--A-ME564-3	400	.56	G, J, K	2.156	.750	2.062	.750
CQ--A-ME684-3	400	.68	G, J, K				
CQ--A-ME824-3	400	.82	G, J, K	2.656	.750	2.562	.750
CQ--A-ME105-3	400	1.0	G, J, K				
CQ--A-MF102-3	600	.0010	G, J, K				
CQ--A-MF122-3	600	.0012	G, J, K				
CQ--A-MF152-3	600	.0015	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF182-3	600	.0018	G, J, K				
CQ--A-MF222-3	600	.0022	G, J, K				
CQ--A-MF272-3	600	.0027	G, J, K				
CQ--A-MF332-3	600	.0033	G, J, K				
CQ--A-MF392-3	600	.0039	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF472-3	600	.0047	G, J, K				
CQ--A-MF562-3	600	.0056	G, J, K				
CQ--A-MF682-3	600	.0068	G, J, K				
CQ--A-MF822-3	600	.0082	G, J, K	1.062	.312	.969	.312 2/

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC M - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MF103-3	600	.010	G, J, K	1.062	.312	.969	.312 2/
CQ--A-MF123-3	600	.012	G, J, K	.938	.400	.844	.400
CQ--A-MF153-3	600	.015	G, J, K				
CQ--A-MF183-3	600	.018	G, J, K	1.062	.400	.969	.400
CQ--A-MF223-3	600	.022	G, J, K				
CQ--A-MF273-3	600	.027	G, J, K	1.312	.400	1.219	.400
CQ--A-MF333-3	600	.033	G, J, K				
CQ--A-MF393-3	600	.039	G, J, K	1.562	.400	1.469	.400
CQ--A-MF473-3	600	.047	G, J, K				
CQ--A-MF563-3	600	.056	G, J, K	1.250	.562	1.156	.562
CQ--A-MF683-3	600	.068	G, J, K				
CQ--A-MF823-3	600	.082	G, J, K	1.500	.562	1.406	.562
CQ--A-MF104-3	600	.10	G, J, K				
CQ--A-MF124-3	600	.12	G, J, K	1.875	.562	1.781	.562
CQ--A-MF154-3	600	.15	G, J, K				
CQ--A-MF184-3	600	.18	G, J, K	1.875	.670	1.781	.670
CQ--A-MF224-3	600	.22	G, J, K				
CQ--A-MF274-3	600	.27	G, J, K	2.125	.750	2.031	.750
CQ--A-MF334-3	600	.33	G, J, K				
CQ--A-MF394-3	600	.39	G, J, K	2.625	.750	2.531	.750
CQ--A-MF474-3	600	.47	G, J, K				
CQ--A-MF564-3	600	.56	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MF684-3	600	.68	G, J, K				
CQ--A-MG102-3	1000	.0010	G, J, K				
CQ--A-MG122-3	1000	.0012	G, J, K				
CQ--A-MG152-3	1000	.0015	G, J, K				
CQ--A-MG182-3	1000	.0018	G, J, K	.938	.312	.844	.312 2/5/
CQ--A-MG222-3	1000	.0022	G, J, K				
CQ--A-MG272-3	1000	.0027	G, J, K				
CQ--A-MG332-3	1000	.0033	G, J, K				
CQ--A-MG392-3	1000	.0039	G, J, K	.938	.400	.844	.400 5/
CQ--A-MG472-3	1000	.0047	G, J, K				
CQ--A-MG562-3	1000	.0056	G, J, K	1.062	.400	.969	.400 5/
CQ--A-MG682-3	1000	.0068	G, J, K				
CQ--A-MG822-3	1000	.0082	G, J, K	1.312	.400	1.219	.400 5/
CQ--A-MG103-3	1000	.010	G, J, K				
CQ--A-MG123-3	1000	.012	G, J, K	1.562	.400	1.469	.400 5/
CQ--A-MG183-3	1000	.018	G, J, K				
CQ--A-MG223-3	1000	.022	G, J, K	1.250	.562	1.156	.562 5/
CQ--A-MG273-3	1000	.027	G, J, K				
CQ--A-MG333-3	1000	.033	G, J, K	1.500	.562	1.406	.562 5/
CQ--A-MG393-3	1000	.039	G, J, K	1.875	.562	1.781	.562 5/
CQ--A-MG473-3	1000	.047	G, J, K				
CQ--A-MG563-3	1000	.056	G, J, K	1.875	.670	1.781	.670 5/
CQ--A-MG683-3	1000	.068	G, J, K				
CQ--A-MG823-3	1000	.082	G, J, K	2.125	.750	2.031	.750
CQ--A-MG104-3	1000	.10	G, J, K				
CQ--A-MG124-3	1000	.12	G, J, K	1.875	1.000	1.781	1.000
CQ--A-MG154-3	1000	.15	G, J, K				
CQ--A-MG184-3	1000	.18	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MG224-3	1000	.22	G, J, K				

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC P				Case dimensions Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-PA102-3	50	.0010	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA122-3	50	.0012	F, J, K				
CQ--A-PA152-3	50	.0015	F, J, K				
CQ--A-PA182-3	50	.0018	F, J, K				
CQ--A-PA222-3	50	.0022	F, J, K				
CQ--A-PA272-3	50	.0027	F, J, K	.875	.235	.812	.235 2/
CQ--A-PA332-3	50	.0033	F, J, K				
CQ--A-PA392-3	50	.0039	F, J, K				
CQ--A-PA472-3	50	.0047	F, J, K				
CQ--A-PA562-3	50	.0056	F, J, K				
CQ--A-PA682-3	50	.0068	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA822-3	50	.0082	F, J, K				
CQ--A-PA103-3	50	.010	F, J, K				
CQ--A-PA123-3	50	.012	F, J, K				
CQ--A-PA153-3	50	.015	F, J, K				
CQ--A-PA183-3	50	.018	F, J, K	.875	.400	.812	.400
CQ--A-PA223-3	50	.022	F, J, K				
CQ--A-PA273-3	50	.027	F, J, K	1.125	.400	1.062	.400
CQ--A-PA333-3	50	.033	F, J, K				
CQ--A-PA393-3	50	.039	F, J, K				
CQ--A-PA473-3	50	.047	F, J, K	1.375	.400	1.312	.400
CQ--A-PA563-3	50	.056	F, J, K				
CQ--A-PA683-3	50	.068	F, J, K	1.125	.562	1.062	.562
CQ--A-PA823-3	50	.082	F, J, K				
CQ--A-PA104-3	50	.10	F, J, K	1.375	.562	1.312	.562
CQ--A-PA124-3	50	.12	F, J, K				
CQ--A-PA154-3	50	.15	F, J, K	1.625	.562	1.562	.562
CQ--A-PA184-3	50	.18	F, J, K				
CQ--A-PA224-3	50	.22	F, J, K	1.625	.670	1.562	.670
CQ--A-PA274-3	50	.27	F, J, K				
CQ--A-PA334-3	50	.33	F, J, K				
CQ--A-PA394-3	50	.39	F, J, K				
CQ--A-PA474-3	50	.47	F, J, K	2.125	.750	2.062	.750
CQ--A-PA564-3	50	.56	F, J, K				
CQ--A-PA684-3	50	.68	F, J, K				
CQ--A-PA824-3	50	.82	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PA105-3	50	1.0	F, J, K				
CQ--A-PC222-3	200	.0022	F, J, K	.875	.235	.812	.235 2/
CQ--A-PC332-3	200	.0033	F, J, K	.875	.312	.812	.312 2/
CQ--A-PC472-3	200	.0047	F, J, K				
CQ--A-PC682-3	200	.0068	F, J, K	.875	.400	.812	.400
CQ--A-PC103-3	200	.010	F, J, K	1.125	.400	1.062	.400
CQ--A-PC153-3	200	.015	F, J, K				
CQ--A-PC223-3	200	.022	F, J, K	1.375	.400	1.312	.400
CQ--A-PC333-3	200	.033	F, J, K	1.125	.562	1.062	.562
CQ--A-PC473-3	200	.047	F, J, K	1.375	.562	1.312	.562
CQ--A-PC683-3	200	.068	F, J, K	1.625	.562	1.562	.562

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

CHARACTERISTIC P - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-PC104-3	200	.10	F, J, K	1.625	.670	1.562	.670
CQ--A-PC154-3	200	.15	F, J, K	2.125	.750	2.062	.750
CQ--A-PC224-3	200	.22	F, J, K				
CQ--A-PC334-3	200	.33	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PC474-3	200	.47	F, J, K				
CQ--A-PE102-3	400	.0010	F, J, K	.875	.235	.812	.235 2/
CQ--A-PE152-3	400	.0015	F, J, K				
CQ--A-PE222-3	400	.0022	F, J, K	.875	.312	.812	.312 2/
CQ--A-PE332-3	400	.0033	F, J, K				
CQ--A-PE472-3	400	.0047	F, J, K	.875	.400	.812	.400
CQ--A-PE682-3	400	.0068	F, J, K	1.125	.400	1.062	.400
CQ--A-PE103-3	400	.010	F, J, K				
CQ--A-PE153-3	400	.015	F, J, K	1.375	.400	1.312	.400
CQ--A-PE223-3	400	.022	F, J, K	1.125	.562	1.062	.562
CQ--A-PE333-3	400	.033	F, J, K	1.375	.562	1.312	.562
CQ--A-PE473-3	400	.047	F, J, K	1.625	.562	1.562	.562
CQ--A-PE683-3	400	.068	F, J, K	1.625	.670	1.562	.670
CQ--A-PE104-3	400	.10	F, J, K	2.125	.750	2.062	.750
CQ--A-PE154-3	400	.15	F, J, K				
CQ--A-PF102-3	600	.0010	F, J, K	.875	.400	.812	.400
CQ--A-PF152-3	600	.0015	F, J, K				
CQ--A-PF222-3	600	.0022	F, J, K				
CQ--A-PF332-3	600	.0033	F, J, K	1.125	.400	1.062	.400
CQ--A-PF472-3	600	.0047	F, J, K				
CQ--A-PF682-3	600	.0068	F, J, K				
CQ--A-PF103-3	600	.010	F, J, K	1.375	.400	1.312	.400
CQ--A-PF153-3	600	.015	F, J, K				
CQ--A-PF223-3	600	.022	F, J, K	1.375	.562	1.312	.562
CQ--A-PF333-3	600	.033	F, J, K	1.375	.562	1.312	.562
CQ--A-PF473-3	600	.047	F, J, K	1.625	.670	1.562	.670
CQ--A-PF683-3	600	.068	F, J, K	2.125	.750	2.062	.750
CQ--A-PF104-3	600	.10	F, J, K				
CQ--A-PF154-3	600	.15	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PF224-3	600	.22	F, J, K	2.625	1.000	2.562	1.000

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors 6/ - Continued.

CHARACTERISTIC T - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 4/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-TC152-3	200	.0015	G, J, K	.812	.235	.719	.235 2/
CQ--A-TC392-3	200	.0039	G, J, K	.938	.312	.844	.312 2/
CQ--A-TC472-3	200	.0047	G, J, K				
CQ--A-TC822-3	200	.0082	G, J, K	.938	.400	.844	.400
CQ--A-TC103-3	200	.010	G, J, K				
CQ--A-TC123-3	200	.012	G, J, K	1.188	.400	1.094	.400
CQ--A-TC153-3	200	.015	G, J, K				
CQ--A-TC183-3	200	.018	G, J, K	1.312	.400	1.219	.400
CQ--A-TC223-3	200	.022	G, J, K				
CQ--A-TC273-3	200	.027	G, J, K	1.125	.562	1.062	.562
CQ--A-TC333-3	200	.033	G, J, K				
CQ--A-TC393-3	200	.039	G, J, K	1.250	.562	1.188	.562
CQ--A-TC473-3	200	.047	G, J, K				
CQ--A-TC563-3	200	.056	G, J, K	1.500	.562	1.438	.562
CQ--A-TC683-3	200	.068	G, J, K				
CQ--A-TC823-3	200	.082	G, J, K	1.875	.562	1.812	.562
CQ--A-TC104-3	200	.10	G, J, K				
CQ--A-TE104-3	400	.10	G, J, K	1.875	.670	1.812	.670
CQ--A-TE124-3	400	.12	G, J, K	2.125	.750	2.062	.750
CQ--A-TE154-3	400	.15	G, J, K				
CQ--A-TE184-3	400	.18	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TE224-3	400	.22	G, J, K				
CQ--A-TE274-3	400	.27	G, J, K	2.375	1.000	2.312	1.000
CQ--A-TE334-3	400	.33	G, J, K				
CQ--A-TF102-3	600	.0010	G, J, K	.812	.235	.719	.235 2/
CQ--A-TF122-3	600	.0012	G, J, K				
CQ--A-TF152-3	600	.0015	G, J, K				
CQ--A-TF182-3	600	.0018	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF222-3	600	.0022	G, J, K				
CQ--A-TF272-3	600	.0027	G, J, K				
CQ--A-TF332-3	600	.0033	G, J, K				
CQ--A-TF392-3	600	.0039	G, J, K				
CQ--A-TF472-3	600	.0047	G, J, K	.938	.400	.844	.400
CQ--A-TF562-3	600	.0056	G, J, K				
CQ--A-TF682-3	600	.0068	G, J, K				
CQ--A-TF822-3	600	.0082	G, J, K	1.188	.400	1.094	.400
CQ--A-TF103-3	600	.010	G, J, K				
CQ--A-TF123-3	600	.012	G, J, K	1.312	.400	1.219	.400
CQ--A-TF153-3	600	.015	G, J, K				
CQ--A-TF183-3	600	.018	G, J, K	1.125	.562	1.062	.562
CQ--A-TF223-3	600	.022	G, J, K				
CQ--A-TF273-3	600	.027	G, J, K	1.250	.562	1.188	.562
CQ--A-TF333-3	600	.033	G, J, K				
CQ--A-TF393-3	600	.039	G, J, K	1.500	.562	1.438	.562
CQ--A-TF473-3	600	.047	G, J, K				

See footnotes at end of table.

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excluded). JAN marking

TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors 6/ - Continued.

CHARACTERISTIC T - Continued				Case dimensions			
				Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 4/)			
				Circuit 1		Circuit 3	
Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-TF563-3	600	.056	G, J, K	1.875	.562	1.812	.562
CQ--A-TF683-3	600	.068	G, J, K				
CQ--A-TF823-3	600	.082	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TF104-3	600	.10	G, J, K				
CQ--A-TF124-3	600	.12	G, J, K	1.875	1.000	2.312	1.000
CQ--A-TF154-3	600	.15	G, J, K				

1/ Complete part number shall include additional symbols to complete style designation, and to indicate circuit, and capacitance tolerance, as applicable.
2/ The minimum D dimension for style CQ13 shall be .400 inch.
3/ For insulated style CQ09 in characteristics K, M, and P add .062 +.062, -.093 inch and .032 +.015, -.037 inch to the nominal L and D dimensions respectively. In no case shall the bare case dimensions of the insulated style be less than the uninsulated minimum.
4/ For insulated style CQ09 in characteristic T add .062 +.062, -.093 inch and .062 +.015, -.068 inch to the nominal L and D dimensions respectively. In no case shall the bare case dimensions of the insulated style be less than the uninsulated minimum.
5/ These units are not available in style CQ13 due to possible arcing with 1,000 volts impressed across the smaller screw necks.
6/ Styles CQ08, CQ09, CQ12 and CQ13 replace and are directly interchangeable with former styles CQ04, CQ05, CQ10 and CQ11 respectively of the same characteristic, voltage, tolerance, etc.

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TABLE IV. Millimeter eqivalents of decimal inches.

Inches	MM	Inches	MM
.005	.13	1.094	27.79
.015	.38	1.125	28.58
.031	.79	1.156	29.36
.175	4.45	1.188	30.18
.195	4.95	1.219	30.96
.235	5.97	1.250	31.75
.312	7.92	1.281	32.54
.400	10.16	1.312	33.32
.406	10.31	1.375	34.93
.468	11.89	1.406	35.71
.500	12.70	1.438	36.53
.562	14.27	1.469	37.31
.625	15.88	1.500	38.10
.656	16.66	1.531	38.89
.670	17.02	1.562	39.67
.687	17.45	1.625	41.28
.688	17.48	1.781	45.24
.718	18.24	1.812	46.02
.719	18.26	1.875	47.63
.749	19.02	1.906	48.41
.750	19.05	2.031	51 59
.781	19.84	2.062	52.37
.812	20.62	2.125	53.98
.844	21.44	2.156	54 76
.875	22.23	2.281	57.94
.906	23.01	2.312	58.72
.938	23.83	2.375	60.33
.969	24.61	2.531	64.29
1.000	25.40	2.562	65.07
1.031	26.10	2.625	66.68
1.062	26.97	2.656	67.46

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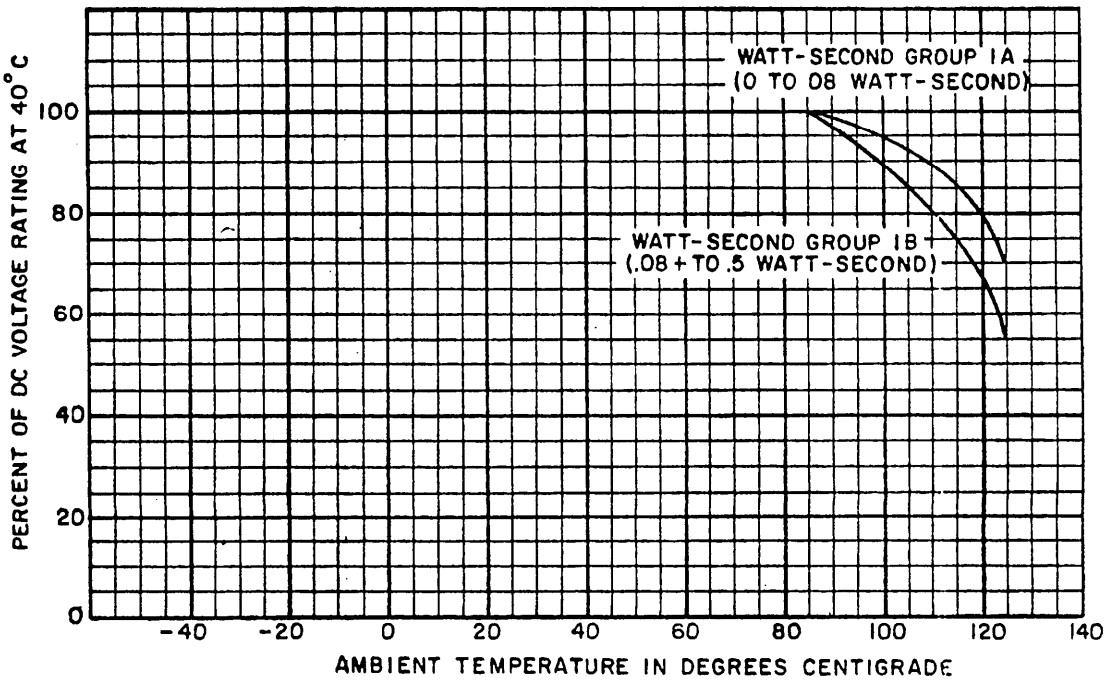


FIGURE 2. Voltage derating for high ambient test temperature (characteristic K).

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TABLE V. Substitutability data.

Superseding part number 1/ MIL-C-19978/9,10,11 Characteristic "K" Styles: CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/ Dash numbers								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KC392-3-	0001-	0002-	0053-	0054-	0105-	0106-	0157-	0158-	CQ--A-KC392-3
CQR--A-KC472-3-	0003-	0004-	0055-	0056-	0107-	0108-	0159-	0160-	CQ--A-KC472-3
CQR--A-KC562-3-	0005-	0006-	0057-	0058-	0109-	0110-	0161-	0162-	CQ--A-KC562-3
CQR--A-KC682-3-	0007-	0008-	0059-	0060-	0111-	0112-	0163-	0164-	CQ--A-KC682-3
CQR--A-KC183-3-	0009-	0010-	0061-	0062-	0113-	0114-	0165-	0166-	CQ--A-KC183-3
CQR--A-KC223-3-	0011-	0012-	0063-	0064-	0115-	0116-	0167-	0168-	CQ--A-KC223-3
CQR--A-KC273-3-	0013-	0014-	0065-	0066-	0117-	0118-	0169-	0170-	CQ--A-KC273-3
CQR--A-KC333-3-	0015-	0016-	0067-	0068-	0119-	0120-	0171-	0172-	CQ--A-KC333-3
CQR--A-KC393-3-	0017-	0018-	0069-	0070-	0121-	0122-	0173-	0174-	CQ--A-KC393-3
CQR--A-KC473-3-	0019-	0020-	0071-	0072-	0123-	0124-	0175-	0176-	CQ--A-KC473-3
CQR--A-KC563-3-	0021-	0022-	0073-	0074-	0125-	0126-	0177-	0178-	CQ--A-KC563-3
CQR--A-KC683-3-	0023-	0024-	0075-	0076-	0127-	0128-	0179-	0180-	CQ--A-KC683-3
CQR--A-KC823-3-	0025-	0026-	0077-	0078-	0129-	0130-	0181-	0182-	CQ--A-KC823-3
CQR--A-KC104-3-	0027-	0028-	0079-	0080-	0131-	0132-	0183-	0184-	CQ--A-KC104-3
CQR--A-KC124-3-	0029-	0030-	0081-	0082-	0133-	0134-	0185-	0186-	CQ--A-KC124-3
CQR--A-KC154-3-	0031-	0032-	0083-	0084-	0135-	0136-	0187-	0188-	CQ--A-KC154-3
CQR--A-KC184-3-	0033-	0034-	0085-	0086-	0137-	0138-	0189-	0190-	CQ--A-KC184-3
CQR--A-KC224-3-	0035-	0036-	0087-	0088-	0139-	0140-	0191-	0192-	CQ--A-KC224-3
CQR--A-KC274-3-	0037-	0038-	0089-	0090-	0141-	0142-	0193-	0194-	CQ--A-KC274-3
CQR--A-KC334-3-	0039-	0040-	0091-	0092-	0143-	0144-	0195-	0196-	CQ--A-KC334-3
CQR--A-KC394-3-	0041-	0042-	0093-	0094-	0145-	0146-	0197-	0198-	CQ--A-KC394-3
CQR--A-KC474-3-	0043-	0044-	0095-	0096-	0147-	0148-	0199-	0200-	CQ--A-KC474-3
CQR--A-KC564-3-	0045-	0046-	0097-	0098-	0149-	0150-	0201-	0202-	CQ--A-KC564-3
CQR--A-KC684-3-	0047-	0048-	0099-	0100-	0151-	0152-	0203-	0204-	CQ--A-KC684-3
CQR--A-KC824-3-	0049-	0050-	0101-	0102-	0153-	0154-	0205-	0206-	CQ--A-KC824-3
CQR--A-KC105-3-	0051-	0052	0103-	0104-	0155-	0156-	0207-	0208-	CQ--A-KC105-3
CQR--A-KE272-3-	0301-	0302-	0341-	0342-	0381-	0382-	0421-	0422-	CQ--A-KE272-3
CQR--A-KE332-3-	0303-	0304-	0343-	0344-	0383-	0384-	0423-	0424-	CQ--A-KE332-3
CQR--A-KE123-3-	0305-	0306-	0345-	0346-	0385-	0386-	0425-	0426-	CQ--A-KE123-3
CQR--A-KE153-3-	0307-	0308-	0347-	0348-	0387-	0388-	0427-	0428-	CQ--A-KE153-3
CQR--A-KE273-3-	0309-	0310-	0349-	0350-	0389-	0390-	0429-	0430-	CQ--A-KE273-3
CQR--A-KE333-3-	0311-	0312-	0351-	0352-	0391-	0392-	0431-	0432-	CQ--A-KE333-3
CQR--A-KE393-3-	0313-	0314-	0353-	0354-	0393-	0394-	0433-	0434-	CQ--A-KE393-3
CQR--A-KE473-3-	0315-	0316-	0355-	0356-	0395-	0396-	0435-	0436-	CQ--A-KE473-3
CQR--A-KE563-3-	0317-	0318-	0357-	0358-	0397-	0398-	0437-	0438-	CQ--A-KE563-3
CQR--A-KE683-3-	0319-	0320-	0359-	0360-	0399-	0400-	0439-	0440-	CQ--A-KE683 3
CQR--A-KE823-3-	0321-	0322-	0361-	0362-	0401-	0402-	0441-	0442-	CQ--A-KE823-3
CQR--A-KE104-3-	0323-	0324-	0363-	0364-	0403-	0404-	0443-	0444-	CQ--A-KE104-3
CQR--A-KE124-3-	0325-	0326-	0365-	0366-	0405-	0406-	0445-	0446-	CQ--A-KE124-3
CQR--A-KE154-3-	0327-	0328-	0367-	0368-	0407-	0408-	0447-	0448-	CQ--A-KE154-3
CQR--A-KE184-3-	0329-	0330-	0369-	0370-	0409-	0410-	0449-	0450-	CQ--A-KE184-3
CQR--A-KE224-3-	0331-	0332-	0371-	0372-	0411-	0412-	0451-	0452-	CQ--A-KE224-3
CQR--A-KE274-3-	0333-	0334-	0373-	0374-	0413-	0414-	0453-	0454-	CQ--A-KE274-3
CQR--A-KE334-3-	0335-	0336-	0375-	0376-	0415-	0416-	0455-	0456-	CQ--A-KE334-3
CQR--A-KE394-3-	0337-	0338-	0377-	0378-	0417-	0418-	0457-	0458-	CQ--A-KE394-3
CQR--A-KE474-3-	0339-	0340-	0379-	0380-	0419-	0420-	0459-	0460-	CQ--A-KE474-3
CQR--A-KF102-3-	0501-	0502-	0567-	0568-	0633-	0634-	0699-	0700-	CQ--A-KF102-3
CQR--A-KF122-3-	0503-	0504-	0569-	0570-	0635-	0636-	0701-	0702-	CQ--A-KF122-3
CQR--A-KF152-3-	0505-	0506-	0571-	0572-	0637-	0638-	0703-	0704-	CQ--A-KF152-3
CQR--A-KF182-3-	0507-	0508-	0573-	0574-	0639-	0640-	0705-	0706-	CQ--A-KF182-3
CQR--A-KF222-3-	0509-	0510-	0575-	0576-	0641-	0642-	0707-	0708-	CQ--A-KF222-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/9,10,11 Characteristic "K" Styles: CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Dash numbers								
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KF272-3-	0511-	0512-	0577-	0578-	0643-	0644-	0709-	0710-	CQ--A-KF272-3
CQR--A-KF332-3-	0513-	0514-	0579-	0580-	0645-	0646-	0711-	0712-	CQ--A-KF332-3
CQR--A-KF392-3-	0515-	0516-	0581-	0582-	0647-	0648-	0713-	0714-	CQ--A-KF392-3
CQR--A-KF472-3-	0517-	0518-	0583-	0584-	0649-	0650-	0715-	0716-	CQ--A-KF472-3
CQR--A-KF562-3-	0519-	0520-	0585-	0586-	0651-	0652-	0717-	0718-	CQ--A-KF562-3
CQR--A-KF682-3-	0521-	0522-	0587-	0588-	0653-	0654-	0719-	0720-	CQ--A-KF682-3
CQR--A-KF822-3-	0523-	0524-	0589-	0590-	0655-	0656-	0721-	0722-	CQ--A-KF822-3
CQR--A-KF103-3-	0525-	0526-	0591-	0592-	0657-	0658-	0723-	0724-	CQ--A-KF103-3
CQR--A-KF123-3-	0527-	0528-	0593-	0594-	0659-	0660-	0725-	0726-	CQ--A-KF123-3
CQR--A-KF153-3-	0529-	0530-	0595-	0596-	0661-	0662-	0727-	0728-	CQ--A-KF153-3
CQR--A-KF183-3-	0531-	0532-	0597-	0598-	0663-	0664-	0729-	0730-	CQ--A-KF183-3
CQR--A-KF223-3-	0533-	0534-	0599-	0600-	0665-	0666-	0731-	0732-	CQ--A-KF223-3
CQR--A-KF273-3-	0535-	0536-	0601-	0602-	0667-	0668-	0733-	0734-	CQ--A-KF273-3
CQR--A-KF333-3-	0537-	0538-	0603-	0604-	0669-	0670-	0735-	0736-	CQ--A-KF333-3
CQR--A-KF393-3-	0539-	0540-	0605-	0606-	0671-	0672-	0737-	0738-	CQ--A-KF393-3
CQR--A-KF473-3-	0541-	0542-	0607-	0608-	0673-	0674-	0739-	0740-	CQ--A-KF473-3
CQR--A-KF563-3-	0543-	0544-	0609-	0610-	0675-	0676-	0741-	0742-	CQ--A-KF563-3
CQR--A-KF683-3-	0545-	0546-	0611-	0612-	0677-	0678-	0743-	0744-	CQ--A-KF683-3
CQR--A-KF823-3-	0547-	0548-	0613-	0614-	0679-	0680-	0745-	0746-	CQ--A-KF823-3
CQR--A-KF104-3-	0549-	0550-	0615-	0616-	0681-	0682-	0747-	0748-	CQ--A-KF104-3
CQR--A-KF124-3-	0551-	0552-	0617-	0618-	0683-	0684-	0749-	0750-	CQ--A-KF124-3
CQR--A-KF154-3-	0553-	0554-	0619-	0620-	0685-	0686-	0751-	0752-	CQ--A-KF154-3
CQR--A-KF184-3-	0555-	0556-	0621-	0622-	0687-	0688-	0753-	0754-	CQ--A-KF184-3
CQR--A-KF224-3-	0557-	0558-	0623-	0624-	0689-	0690-	0755-	0756-	CQ--A-KF224-3
CQR--A-KF274-3-	0559-	0560-	0625-	0626-	0691-	0692-	0757-	0758-	CQ--A-KF274-3
CQR--A-KF334-3-	0561-	0562-	0627-	0628-	0693-	0694-	0759-	0760-	CQ--A-KF334-3
CQR--A-KF394-3-	0563-	0564-	0629-	0630-	0695-	0696-	0761-	0762-	CQ--A-KF394-3
CQR--A-KF474-3-	0565-	0566-	0631-	0632-	0697-	0698-	0763-	0764-	CQ--A-KF474-3
CQR--A-KG102-3-	0801-	0802-	0867-	0868-	0933-	0934-	0999-	1000-	CQ--A-KG102-3
CQR--A-KG122-3-	0803-	0804-	0869-	0870-	0935-	0936-	1001-	1002-	CQ--A-KG122-3
CQR--A-KG152-3-	0805-	0806-	0871-	0872-	0937-	0938-	1003-	1004-	CQ--A-KG152-3
CQR--A-KG182-3-	0807-	0808-	0873-	0874-	0939-	0940-	1005-	1006-	CQ--A-KG182-3
CQR--A-KG222-3-	0809-	0810-	0875-	0876-	0941-	0942-	1007-	1008-	CQ--A-KG222-3
CQR--A-KG272-3-	0811-	0812-	0877-	0878-	0943-	0944-	1009-	1010-	CQ--A-KG272-3
CQR--A-KG332-3-	0813-	0814-	0879-	0880-	0945-	0946-	1011-	1012-	CQ--A-KG332-3
CQR--A-KG392-3-	0815-	0816-	0881-	0882-	0947-	0948-	1013-	1014-	CQ--A-KG392-3
CQR--A-KG472-3-	0817-	0818-	0883-	0884-	0949-	0950-	1015-	1016-	CQ--A-KG472-3
CQR--A-KG562-3-	0819-	0820-	0885-	0886-	0951-	0952-	1017-	1018-	CQ--A-KG562-3
CQR--A-KG682-3-	0821-	0822-	0887-	0888-	0953-	0954-	1019-	1020-	CQ--A-KG682-3
CQR--A-KG822-3-	0823-	0824-	0889-	0890-	0955-	0956-	1021-	1022-	CQ--A-KG822-3
CQR--A-KG103-3-	0825-	0826-	0891-	0892-	0957-	0958-	1023-	1024-	CQ--A-KG103-3
CQR--A-KG123-3-	0827-	0828-	0893-	0894-	0959-	0960-	1025-	1026-	CQ--A-KG123-3
CQR--A-KG153-3-	0829-	0830-	0895-	0896-	0961-	0962-	1027-	1028-	CQ--A-KG153-3
CQR--A-KG183-3-	0831-	0832-	0897-	0898-	0963-	0964-	1029-	1030-	CQ--A-KG183-3
CQR--A-KG223-3-	0833-	0834-	0899-	0900-	0965-	0966-	1031-	1032-	CQ--A-KG223-3
CQR--A-KG273-3-	0835-	0836-	0901-	0902-	0967-	0968-	1033-	1034-	CQ--A-KG273-3
CQR--A-KG333-3-	0837-	0838-	0903-	0904-	0969-	0970-	1035-	1036-	CQ--A-KG333-3
CQR--A-KG393-3-	0839-	0840-	0905-	0906-	0971-	0972-	1037-	1038-	CQ--A-KG393-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/9,10,11 Characteristic "K" Styles: CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Dash numbers								
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KG473-3-	0841-	0842-	0907-	0908-	0973-	0974-	1039-	1040-	CQ--A-KG473-3
CQR--A-KG563-3-	0843-	0844-	0909-	0910-	0975-	0976-	1041-	1042-	CQ--A-KG563-3
CQR--A-KG683-3-	0845-	0846-	0911-	0912-	0977-	0978-	1043-	1044-	CQ--A-KG683-3
CQR--A-KG823-3-	0847-	0848-	0913-	0914-	0979-	0980-	1045-	1046-	CQ--A-KG823-3
CQR--A-KG104-3-	0849-	0850-	0915-	0916-	0981-	0982-	1047-	1048-	CQ--A-KG104-3
CQR--A-KG124-3-	0851-	0852-	0917-	0918-	0983-	0984-	1049-	1050-	CQ--A-KG124-3
CQR--A-KG154-3-	0853-	0854-	0919-	0920-	0985-	0986-	1051-	1052-	CQ--A-KG154-3
CQR--A-KG184-3-	0855-	0856-	0921-	0922-	0987-	0988-	1053-	1054-	CQ--A-KG184-3
CQR--A-KG224-3-	0857-	0858-	0923-	0924-	0989-	0990-	1055-	1056-	CQ--A-KG224-3
CQR--A-KG274-3-	0859-	0860-	0925-	0926-	0991-	0992-	1057-	1058-	CQ--A-KG274-3
CQR--A-KG334-3-	0861-	0862-	0927-	0928-	0993-	0994-	1059-	1060-	CQ--A-KG334-3
CQR--A-KG394-3-	0863-	0864-	0929-	0930-	0995-	0996-	1061-	1062-	CQ--A-KG394-3
CQR--A-KG474-3-	0865-	0866-	0931-	0932-	0997-	0998-	1063-	1064-	CQ--A-KG474-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/13,14,15 Characteristic "M" Styles: CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/ Dash numbers								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MZ155-3-	1101-	1102-	1113-	1114-	1125-	1126-	1137-	1138-	CQ--A-MZ155-3
CQR--A-MZ225-3-	1103-	1104-	1115-	1116-	1127-	1128-	1139-	1140-	CQ--A-MZ225-3
CQR--A-MZ335-3-	1105-	1106-	1117-	1118-	1129-	1130-	1141-	1142-	CQ--A-MZ335-3
CQR--A-MZ475-3-	1107-	1108-	1119-	1120-	1131-	1132-	1143-	1144-	CQ--A-MZ475-3
CQR--A-MZ685-3-	1109-	1110-	1121-	1122-	1133-	1134-	1145-	1146-	CQ--A-MZ685-3
CQR--A-MZ106-3-	1111-	1112-	1123-	1124-	1135-	1136-	1147-	1148-	CQ--A-MZ106-3
CQR--A-MA102-3-	1201-	1202-	1273-	1274-	1345-	1346-	1417-	1418-	CQ--A-MA102-3
CQR--A-MA122-3-	1203-	1204-	1275-	1276-	1347-	1348-	1419-	1420-	CQ--A-MA122-3
CQR--A-MA152-3-	1205-	1206-	1277-	1278-	1349-	1350-	1421-	1422-	CQ--A-MA152-3
CQR--A-MA182-3-	1207-	1208-	1279-	1280-	1351-	1352-	1423-	1424-	CQ--A-MA182-3
CQR--A-MA222-3-	1209-	1210-	1281-	1282-	1353-	1354-	1425-	1426-	CQ--A-MA222-3
CQR--A-MA272-3-	1211-	1212-	1283-	1284-	1355-	1356-	1427-	1428-	CQ--A-MA272-3
CQR--A-MA332-3-	1213-	1214-	1285-	1286-	1357-	1358-	1429-	1430-	CQ--A-MA332-3
CQR--A-MA392-3-	1215-	1216-	1287-	1288-	1359-	1360-	1431-	1432-	CQ--A-MA392-3
CQR--A-MA472-3-	1217-	1218-	1289-	1290-	1361-	1362-	1433-	1434-	CQ--A-MA472-3
CQR--A-MA562-3-	1219-	1220-	1291-	1292-	1363-	1364-	1435-	1436-	CQ--A-MA562-3
CQR--A-MA682-3-	1221-	1222-	1293-	1294-	1365-	1366-	1437-	1438-	CQ--A-MA682-3
CQR--A-MA822-3-	1223-	1224-	1295-	1296-	1367-	1368-	1439-	1440-	CQ--A-MA822-3
CQR--A-MA103-3-	1225-	1226-	1297-	1298-	1369-	1370-	1441-	1442-	CQ--A-MA103-3
CQR--A-MA123-3-	1227-	1228-	1299-	1300-	1371-	1372-	1443-	1444-	CQ--A-MA123-3
CQR--A-MA153-3-	1229-	1230-	1301-	1302-	1373-	1374-	1445-	1446-	CQ--A-MA153-3
CQR--A-MA183-3-	1231-	1232-	1303-	1304-	1375-	1376-	1447-	1448-	CQ--A-MA183-3
CQR--A-MA223-3-	1233-	1234-	1305-	1306-	1377-	1378-	1449-	1450-	CQ--A-MA223-3
CQR--A-MA273-3-	1235-	1236-	1307-	1308-	1379-	1380-	1451-	1452-	CQ--A-MA273-3
CQR--A-MA333-3-	1237-	1238-	1309-	1310-	1381-	1382-	1453-	1454-	CQ--A-MA333-3
CQR--A-MA393-3-	1239-	1240-	1311-	1312-	1383-	1384-	1455-	1456-	CQ--A-MA393-3
CQR--A-MA473-3-	1241-	1242-	1313-	1314-	1385-	1386-	1457-	1458-	CQ--A-MA473-3
CQR--A-MA563-3-	1243-	1244-	1315-	1316-	1387-	1388-	1459-	1460-	CQ--A-MA563-3
CQR--A-MA683-3-	1245-	1246-	1317-	1318-	1389-	1390-	1461-	1462-	CQ--A-MA683-3
CQR--A-MA124-3-	1247-	1248-	1319-	1320-	1391-	1392-	1463-	1464-	CQ--A-MA124-3
CQR--A-MA154-3-	1249-	1250-	1321-	1322-	1393-	1394-	1465-	1466-	CQ--A-MA154-3
CQR--A-MA184-3-	1251-	1252-	1323-	1324-	1395-	1396-	1467-	1468-	CQ--A-MA184-3
CQR--A-MA224-3-	1253-	1254-	1325-	1326-	1397-	1398-	1469-	1470-	CQ--A-MA224-3
CQR--A-MA274-3-	1255-	1256-	1327-	1328-	1399-	1400-	1471-	1472-	CQ--A-MA274-3
CQR--A-MA334-3-	1257-	1258-	1329-	1330-	1401-	1402-	1473-	1474-	CQ--A-MA334-3
CQR--A-MA394-3-	1259-	1260-	1331-	1332-	1403-	1404-	1475-	1476-	CQ--A-MA394-3
CQR--A-MA474-3-	1261-	1262-	1333-	1334-	1405-	1406-	1477-	1478-	CQ--A-MA474-3
CQR--A-MA564-3-	1263-	1264-	1335-	1336-	1407-	1408-	1479-	1480-	CQ--A-MA564-3
CQR--A-MA684-3-	1265-	1266-	1337-	1338-	1409-	1410-	1481-	1482-	CQ--A-MA684-3
CQR--A-MA824-3-	1267-	1268-	1339-	1340-	1411-	1412-	1483-	1484-	CQ--A-MA824-3
CQR--A-MA105-3-	1269-	1270-	1341-	1342-	1413-	1414-	1485-	1486-	CQ--A-MA105-3
CQR--A-MA125-3-	1271-	1272-	1343-	1344-	1415-	1416-	1487-	1488-	CQ--A-MA125-3
CQR--A-MB125-3-	1501-	1502-	1521-	1522-	1541-	1542-	1561-	1562-	CQ--A-MB125-3
CQR--A-MB155-3-	1503-	1504-	1523-	1524-	1543-	1544-	1563-	1564-	CQ--A-MB155-3
CQR--A-MB185-3-	1505-	1506-	1525-	1526-	1545-	1546-	1565-	1566-	CQ--A-MB185-3
CQR--A-MB225-3-	1507-	1508-	1527-	1528-	1547-	1548-	1567-	1568-	CQ--A-MB225-3
CQR--A-MB275-3-	1509-	1510-	1529-	1530-	1549-	1550-	1569-	1570-	CQ--A-MB275-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/13,14,15 Characteristic "M" Styles: CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Dash numbers								
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MB335-3-	1511-	1512-	1531-	1532-	1551-	1552-	1571-	1572-	CQ--A-MB335-3
CQR--A-MB395-3-	1513-	1514-	1533-	1534-	1553-	1554-	1573-	1574-	CQ--A-MB395-3
CQR--A-MB475-3-	1515-	1516-	1535-	1536-	1555-	1556-	1575-	1576-	CQ--A-MB475-3
CQR--A-MB565-3-	1517-	1518-	1537-	1538-	1557-	1558-	1577-	1578-	CQ--A-MB565-3
CQR--A-MB685-3-	1519-	1520-	1539-	1540-	1559-	1560-	1579-	1580-	CQ--A-MB685-3
CQR--A-MC102-3-	1601-	1602-	1675-	1676-	1749-	1750-	1823-	1824-	CQ--A-MC102-3
CQR--A-MC122-3-	1603-	1604-	1677-	1678-	1751-	1752-	1825-	1826-	CQ--A-MC122-3
CQR--A-MC152-3-	1605-	1606-	1679-	1680-	1753-	1754-	1827-	1828-	CQ--A-MC152-3
CQR--A-MC182-3-	1607-	1608-	1681-	1682-	1755-	1756-	1829-	1830-	CQ--A-MC182-3
CQR--A-MC222-3-	1609-	1610-	1683-	1684-	1757-	1758-	1831-	1832-	CQ--A-MC222-3
CQR--A-MC272-3-	1611-	1612-	1685-	1686-	1759-	1760-	1833-	1834-	CQ--A-MC272-3
CQR--A-MC332-3-	1613-	1614-	1687-	1688-	1761-	1762-	1835-	1836-	CQ--A-MC332-3
CQR--A-MC392-3-	1615-	1616-	1689-	1690-	1763-	1764-	1837-	1838-	CQ--A-MC392-3
CQR--A-MC472-3-	1617-	1618-	1691-	1692-	1765-	1766-	1839-	1840-	CQ--A-MC472-3
CQR--A-MC562-3-	1619-	1620-	1693-	1694-	1767-	1768-	1841-	1842-	CQ--A-MC562-3
CQR--A-MC682-3-	1621-	1622-	1695-	1696-	1769-	1770-	1843-	1844-	CQ--A-MC682-3
CQR--A-MC822-3-	1623-	1624-	1697-	1698-	1771-	1772-	1845-	1846-	CQ--A-MC822-3
CQR--A-MC103-3-	1625-	1626-	1699-	1700-	1773-	1774-	1847-	1848-	CQ--A-MC103-3
CQR--A-MC123-3-	1627-	1628-	1701-	1702-	1775-	1776-	1849-	1850-	CQ--A-MC123-3
CQR--A-MC153-3-	1629-	1630-	1703-	1704-	1777-	1778-	1851-	1852-	CQ--A-MC153-3
CQR--A-MC183-3-	1631-	1632-	1705-	1706-	1779-	1780-	1853-	1854-	CQ--A-MC183-3
CQR--A-MC223-3-	1633-	1634-	1707-	1708-	1781-	1782-	1855-	1856-	CQ--A-MC223-3
CQR--A-MC273-3-	1635-	1636-	1709-	1710-	1783-	1784-	1857-	1858-	CQ--A-MC273-3
CQR--A-MC333-3-	1637-	1638-	1711-	1712-	1785-	1786-	1859-	1860-	CQ--A-MC333-3
CQR--A-MC393-3-	1639-	1640-	1713-	1714-	1787-	1788-	1861-	1862-	CQ--A-MC393-3
CQR--A-MC473-3-	1641-	1642-	1715-	1716-	1789-	1790-	1863-	1864-	CQ--A-MC473-3
CQR--A-MC563-3-	1643-	1644-	1717-	1718-	1791-	1792-	1865-	1866-	CQ--A-MC563-3
CQR--A-MC683-3-	1645-	1646-	1719-	1720-	1793-	1794-	1867-	1868-	CQ--A-MC683-3
CQR--A-MC823-3-	1647-	1648-	1721-	1722-	1795-	1796-	1869-	1870-	CQ--A-MC823-3
CQR--A-MC104-3-	1649-	1650-	1723-	1724-	1797-	1798-	1871-	1872-	CQ--A-MC104-3
CQR--A-MC124-3-	1651-	1652-	1725-	1726-	1799-	1800-	1873-	1874-	CQ--A-MC124-3
CQR--A-MC154-3-	1653-	1654-	1727-	1728-	1801-	1802-	1875-	1876-	CQ--A-MC154-3
CQR--A-MC184-3-	1655-	1656-	1729-	1730-	1803-	1804-	1877-	1878-	CQ--A-MC184-3
CQR--A-MC224-3-	1657-	1658-	1731-	1732-	1805-	1806-	1879-	1880-	CQ--A-MC224-3
CQR--A-MC274-3-	1659-	1660-	1733-	1734-	1807-	1808-	1881-	1882-	CQ--A-MC274-3
CQR--A-MC334-3-	1661-	1662-	1735-	1736-	1809-	1810-	1883-	1884-	CQ--A-MC334-3
CQR--A-MC394-3-	1663-	1664-	1737-	1738-	1811-	1812-	1885-	1886-	CQ--A-MC394-3
CQR--A-MC474-3-	1665-	1666-	1739-	1740-	1813-	1814-	1887-	1888-	CQ--A-MC474-3
CQR--A-MC564-3-	1667-	1668-	1741-	1742-	1815-	1816-	1889-	1890-	CQ--A-MC564-3
CQR--A-MC684-3-	1669-	1670-	1743-	1744-	1817-	1818-	1891-	1892-	CQ--A-MC684-3
CQR--A-MC824-3-	1671-	1672-	1745-	1746-	1819-	1820-	1893-	1894-	CQ--A-MC824-3
CQR--A-MC105-3-	1673-	1674-	1747-	1748-	1821-	1822-	1895-	1896-	CQ--A-MC105-3
CQR--A-ME102-3-	2001-	2002-	2075-	2076-	2149-	2150-	2223-	2224-	CQ--A-ME102-3
CQR--A-ME122-3-	2003-	2004-	2077-	2078-	2151-	2152-	2225-	2226-	CQ--A-ME122-3
CQR--A-ME152-3-	2005-	2006-	2079-	2080-	2153-	2154-	2227-	2228-	CQ--A-ME152-3
CQR--A-ME182-3-	2007-	2008-	2081-	2082-	2155-	2156-	2229-	2230-	CQ--A-ME182-3
CQR--A-ME222-3-	2009-	2010-	2083-	2084-	2157-	2158-	2231-	2232-	CQ--A-ME222-3

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/13,14,15 Characteristic "M" Styles: CQR29 CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D (D)
	Dash numbers								
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-ME272-3-	2011-	2012-	2085-	2086-	2159-	2160-	2233-	2234-	CQ--A-ME272-3
CQR--A-ME332-3-	2013-	2014-	2087-	2088-	2161-	2162-	2235-	2236-	CQ--A-ME332-3
CQR--A-ME392-3-	2015-	2016-	2089-	2090-	2163-	2164-	2237-	2238-	CQ--A-ME392-3
CQR--A-ME472-3-	2017-	2018-	2091-	2092-	2165-	2166-	2239-	2240-	CQ--A-ME472-3
CQR--A-ME562-3-	2019-	2020-	2093-	2094-	2167-	2168-	2241-	2242-	CQ--A-ME562-3
CQR--A-ME682-3-	2021-	2022-	2095-	2096-	2169-	2170-	2243-	2244-	CQ--A-ME682-3
CQR--A-ME822-3-	2023-	2024-	2097-	2098-	2171-	2172-	2245-	2246-	CQ--A-ME822-3
CQR--A-ME103-3-	2025-	2026-	2099-	2100-	2173-	2174-	2247-	2248-	CQ--A-ME103-3
CQR--A-ME123-3-	2027-	2028-	2101-	2102-	2175-	2176-	2249-	2250-	CQ--A-ME123-3
CQR--A-ME153-3-	2029-	2030-	2103-	2104-	2177-	2178-	2251-	2252-	CQ--A-ME153-3
CQR--A-ME183-3-	2031-	2032-	2105-	2106-	2179-	2180-	2253-	2254-	CQ--A-ME183-3
CQR--A-ME223-3-	2033-	2034-	2107-	2108-	2181-	2182-	2255-	2256-	CQ--A-ME223-3
CQR--A-ME273-3-	2035-	2036-	2109-	2110-	2183-	2184-	2257-	2258-	CQ--A-ME273-3
CQR--A-ME333-3-	2037-	2038-	2111-	2112-	2185-	2186-	2259-	2260-	CQ--A-ME333-3
CQR--A-ME393-3-	2039-	2040-	2113-	2114-	2187-	2188-	2261-	2262-	CQ--A-ME393-3
CQR--A-ME473-3-	2041-	2042-	2115-	2116-	2189-	2190-	2263-	2264-	CQ--A-ME473-3
CQR--A-ME563-3-	2043-	2044-	2117-	2118-	2191-	2192-	2265-	2266-	CQ--A-ME563-3
CQR--A-ME683-3-	2045-	2046-	2119-	2120-	2193-	2194-	2267-	2268-	CQ--A-ME683-3
CQR--A-ME823-3-	2047-	2048-	2121-	2122-	2195-	2196-	2269-	2270-	CQ--A-ME823-3
CQR--A-ME104-3-	2049-	2050-	2123-	2124-	2197-	2198-	2271-	2272-	CQ--A-ME104-3
CQR--A-ME124-3-	2051-	2052-	2125-	2126-	2199-	2200-	2273-	2274-	CQ--A-ME124-3
CQR--A-ME154-3-	2053-	2054-	2127-	2128-	2201-	2202-	2275-	2276-	CQ--A-ME154-3
CQR--A-ME184-3-	2055-	2056-	2129-	2130-	2203-	2204-	2277-	2278-	CQ--A-ME184-3
CQR--A-ME224-3-	2057-	2058-	2131-	2132-	2205-	2206-	2279-	2280-	CQ--A-ME224-3
CQR--A-ME274-3-	2059-	2060-	2133-	2134-	2207-	2208-	2281-	2282-	CQ--A-ME274-3
CQR--A-ME334-3-	2061-	2062-	2135-	2136-	2209-	2210-	2283-	2284-	CQ--A-ME334-3
CQR--A-ME394-3-	2063-	2064-	2137-	2138-	2211-	2212-	2285-	2286-	CQ--A-ME394-3
CQR--A-ME474-3-	2065-	2066-	2139-	2440-	2213-	2214-	2287-	2288-	CQ--A-ME474-3
CQR--A-ME564-3-	2067-	2068-	2141-	2442-	2215-	2216-	2289-	2290-	CQ--A-ME564-3
CQR--A-ME684-3-	2069-	2070-	2143-	2144-	2217-	2218-	2291-	2292-	CQ--A-ME684-3
CQR--A-ME824-3-	2071-	2072-	2145-	2146-	2219-	2220-	2293-	2294-	CQ--A-ME824-3
CQR--A-ME105-3-	2073-	2074-	2147-	2148-	2221-	2222-	2295-	2296-	CQ--A-ME105-3
CQR--A-MF102-3-	2401-	2402-	2471-	2472-	2541-	2542-	2611-	2612-	CQ--A-MF102-3
CQR--A-MF122-3-	2403-	2404-	2473-	2474-	2543-	2544-	2613-	2614-	CQ--A-MF122-3
CQR--A-MF152-3-	2405-	2406-	2475-	2476-	2545-	2546-	2615-	2616-	CQ--A-MF152-3
CQR--A-MF182-3-	2407-	2408-	2477-	2478-	2547-	2548-	2617-	2618-	CQ--A-MF182-3
CQR--A-MF222-3-	2409-	2410-	2479-	2480-	2549-	2550-	2619-	2620-	CQ--A-MF222-3
CQR--A-MF272-3-	2411-	2412-	2481-	2482-	2551-	2552-	2621-	2622-	CQ--A-MF272-3
CQR--A-MF332-3-	2413-	2414-	2483-	2484-	2553-	2554-	2623-	2624-	CQ--A-MF332-3
CQR--A-MF392-3-	2415-	2416-	2485-	2486-	2555-	2556-	2625-	2626-	CQ--A-MF392-3
CQR--A-MF472-3-	2417-	2418-	2487-	2488-	2557-	2558-	2627-	2628-	CQ--A-MF472-3
CQR--A-MF562-3-	2419-	2420-	2489-	2490-	2559-	2560-	2629-	2630-	CQ--A-MF562-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-C-19978/13,14,15 Characteristic "M" Styles: CQR29 CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B MIL-C-19978/1C MIL-C-19978/1D
	Dash numbers								
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MF682-3-	2421-	2422-	2491-	2492-	2561-	2562-	2631-	2632-	CQ--A-MF682-3
CQR--A-MF822-3-	2423-	2424-	2493-	2494-	2563-	2564-	2633-	2634-	CQ--A-MF822-3
CQR--A-MF103-3-	2425-	2426-	2495-	2496-	2565-	2566-	2635-	2636-	CQ--A-MF103-3
CQR--A-MF123-3-	2427-	2428-	2497-	2498-	2567-	2568-	2637-	2638-	CQ--A-MF123-3
CQR--A-MF153-3-	2429-	2430-	2499-	2500-	2569-	2570-	2639-	2640-	CQ--A-MF153-3
CQR--A-MF183-3-	2431-	2432-	2501-	2502-	2571-	2572-	2641-	2642-	CQ--A-MF183-3
CQR--A-MF223-3-	2433-	2434-	2503-	2504-	2573-	2574-	2643-	2644-	CQ--A-MF223-3
CQR--A-MF273-3-	2435-	2436-	2505-	2506-	2575-	2576-	2645-	2646-	CQ--A-MF273-3
CQR--A-MF333-3-	2437-	2438-	2507-	2508-	2577-	2578-	2647-	2648-	CQ--A-MF333-3
CQR--A-MF393-3-	2439-	2440-	2509-	2510-	2579-	2580-	2649-	2650-	CQ--A-MF393-3
CQR--A-MF473-3-	2441-	2442-	2511-	2512-	2581-	2582-	2651-	2652-	CQ--A-MF473-3
CQR--A-MF563-3-	2443-	2444-	2513-	2514-	2583-	2584-	2653-	2654-	CQ--A-MF563-3
CQR--A-MF683-3-	2445-	2446-	2515-	2516-	2585-	2586-	2655-	2656-	CQ--A-MF683-3
CQR--A-MF823-3-	2447-	2448-	2517-	2518-	2587-	2588-	2657-	2658-	CQ--A-MF823-3
CQR--A-MF104-3-	2449-	2450-	2519-	2520-	2589-	2590-	2659-	2660-	CQ--A-MF104-3
CQR--A-MF124-3-	2451-	2452-	2521-	2522-	2591-	2592-	2661-	2662-	CQ--A-MF124-3
CQR--A-MF154-3-	2453-	2454-	2523-	2524-	2593-	2594-	2663-	2664-	CQ--A-MF154-3
CQR--A-MF184-3-	2455-	2456-	2525-	2526-	2595-	2596-	2665-	2666-	CQ--A-MF184-3
CQR--A-MF224-3-	2457-	2458-	2527-	2528-	2597-	2598-	2667-	2668-	CQ--A-MF224-3
CQR--A-MF274-3-	2459-	2460-	2529-	2530-	2599-	2600-	2669-	2670-	CQ--A-MF274-3
CQR--A-MF334-3-	2461-	2462-	2531-	2532-	2601-	2602-	2671-	2672-	CQ--A-MF334-3
CQR--A-MF394-3-	2463-	2464-	2533-	2534-	2603-	2604-	2673-	2674-	CQ--A-MF394-3
CQR--A-MF474-3-	2465-	2466-	2535-	2536-	2605-	2606-	2675-	2676-	CQ--A-MF474-3
CQR--A-MF564-3-	2467-	2468-	2537-	2538-	2607-	2608-	2677-	2678-	CQ--A-MF564-3
CQR--A-MF684-3-	2469-	2470-	2539-	2540-	2609-	2610-	2679-	2680-	CQ--A-MF684-3
CQR--A-MG102-3-	2701-	2702-	2757-	2758-	2813-	2814-	2869-	2870-	CQ--A-MG102-3
CQR--A-MG122-3-	2703-	2704-	2759-	2760-	2815-	2816-	2871-	2872-	CQ--A-MG122-3
CQR--A-MG152-3-	2705-	2706-	2761-	2762-	2817-	2818-	2873-	2874-	CQ--A-MG152-3
CQR--A-MG182-3-	2707-	2708-	2763-	2764-	2819-	2820-	2875-	2876-	CQ--A-MG182-3
CQR--A-MG222-3-	2709-	2710-	2765-	2766-	2821-	2822-	2877-	2878-	CQ--A-MG222-3
CQR--A-MG272-3-	2711-	2712-	2767-	2768-	2823-	2824-	2879-	2880-	CQ--A-MG272-3
CQR--A-MG332-3-	2713-	2714-	2769-	2770-	2825-	2826-	2881-	2882-	CQ--A-MG332-3
CQR--A-MG392-3-	2715-	2716-	2771-	2772-	2827-	2828-	2883-	2884-	CQ--A-MG392-3
CQR--A-MG472-3-	2717-	2718-	2773-	2774-	2829-	2830-	2885-	2886-	CQ--A-MG472-3
CQR--A-MG562-3-	2719-	2720-	2775-	2776-	2831-	2832-	2887-	2888-	CQ--A-MG562-3
CQR--A-MG682-3-	2721-	2722-	2777-	2778-	2833-	2834-	2889-	2890-	CQ--A-MG682-3
CQR--A-MG822-3-	2723-	2724-	2779-	2780-	2835-	2836-	2891-	2892-	CQ--A-MG822-3
CQR--A-MG103-3-	2725-	2726-	2781-	2782-	2837-	2838-	2893-	2894-	CQ--A-MG103-3
CQR--A-MG123-3-	2727-	2728-	2783-	2784-	2839-	2840-	2895-	2896-	CQ--A-MG123-3
CQR--A-MG183-3-	2729-	2730-	2785-	2786-	2841-	2842-	2897-	2898-	CQ--A-MG183-3
CQR--A-MG223-3-	2731-	2732-	2787-	2788-	2843-	2844-	2899-	2900-	CQ--A-MG223-3
CQR--A-MG273-3-	2733-	2734-	2789-	2790-	2845-	2846-	2901-	2902-	CQ--A-MG273-3
CQR--A-MG333-3-	2735-	2736-	2791-	2792-	2847-	2848-	2903-	2904-	CQ--A-MG333-3
CQR--A-MG393-3-	2737-	2738-	2793-	2794-	2849-	2850-	2905-	2906-	CQ--A-MG393-3
CQR--A-MG473-3-	2739-	2740-	2795-	2796-	2851-	2852-	2907-	2908-	CQ--A-MG473-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/
MIL-C-19978/13,14,15	Dash numbers								MIL-C-19978/1
Characteristic "M"	Style CQ08		Style CQ09		Style CQ12		Style CQ13		MIL-C-19978/1B
Styles: CQR29	Circuit	Circuit	Circuit	Circuit	Circuit	Circuit	Circuit	Circuit	MIL-C-19978/1C
CQR32, CQR33	1	3	1	3	1	3	1	3	MIL-C-19978/1D
CQR--A-MG563-3-	2741-	2742-	2797-	2798-	2853-	2854-	2909-	2910-	CQ--A-MG563-3
CQR--A-MG683-3-	2743-	2744-	2799-	2800-	2855-	2856-	2911-	2912-	CQ--A-MG683-3
CQR--A-MG823-3-	2745-	2746-	2801-	2802-	2857-	2858-	2913-	2914-	CQ--A-MG823-3
CQR--A-MG104-3-	2747-	2748-	2803-	2804-	2859-	2860-	2915-	2916-	CQ--A-MG104-3
CQR--A-MG124-3-	2749-	2750-	2805-	2806-	2861-	2862-	2917-	2918-	CQ--A-MG124-3
CQR--A-MG154-3-	2751-	2752-	2807-	2808-	2863-	2864-	2919-	2920-	CQ--A-MG154-3
CQR--A-MG184-3-	2753-	2754-	2809-	2810-	2865-	2866-	2921-	2922-	CQ--A-MG184-3
CQR--A-MG224-3-	2755-	2756-	2811-	2812-	2867-	2868-	2923-	2924-	CQ--A-MG224-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ (D)
Dash numbers								MIL-C-19978/1
Style CQ08		Style CQ09		Style CQ12		Style CQ13		MIL-C-19978/1B
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	MIL-C-19978/1C MIL-C-19978/1D
3101-	3102-	3131-	3132-	3161-	3162-	3191-	3192-	CQ--A-PC222-3
3103-	3104-	3133-	3134-	3163-	3164-	3193-	3194-	CQ--A-PC332-3
3105-	3106-	3135-	3136-	3165-	3166-	3195-	3196-	CQ--A-PC472-3
3107-	3108-	3137-	3138-	3167-	3168-	3197-	3198-	CQ--A-PC682-3
3109-	3110-	3139-	3140-	3169-	3170-	3199-	3200-	CQ--A-PC103-3
3111-	3112-	3141-	3142-	3171-	3172-	3201-	3202-	CQ--A-PC153-3
3113-	3114-	3143-	3144-	3173-	3174-	3203-	3204-	CQ--A-PC223-3
3115-	3116-	3145-	3146-	3175-	3176-	3205-	3206-	CQ--A-PC333-3
3117-	3118-	3147-	3148-	3177-	3178-	3207-	3208-	CQ--A-PC473-3
3119-	3120-	3149-	3150-	3179-	3180-	3209-	3210-	CQ--A-PC683 3
3121-	3122-	3151-	3152-	3181-	3182-	3211-	3212-	CQ--A-PC104-3
3123-	3124-	3153-	3154-	3183-	3184-	3213-	3214-	CQ--A-PC154-3
3125-	3126-	3155-	3156-	3185-	3186-	3215-	3216-	CQ--A-PC224-3
3127-	3128-	3157-	3158-	3187-	3188-	3217-	3218-	CQ--A-PC334-3
3129-	3130-	3159-	3160-	3189-	3190-	3219-	3220-	CQ--A-PC474-3
3301-	3302-	3329-	3330-	3357-	3358-	3385-	3386-	CQ--A-PE102-3
3303-	3304-	3331-	3332-	3359-	3360-	3387-	3388-	CQ--A-PE152-3
3305-	3306-	3333-	3334-	3361-	3362-	3389-	3390-	CQ--A-PE222-3
3307-	3308-	3335-	3336-	3363-	3364-	3391-	3392-	CQ--A-PE332-3
3309-	3310-	3337-	3338-	3365-	3366-	3393-	3394-	CQ--A-PE472-3
3311-	3312-	3339-	3340-	3367-	3368-	3395-	3396-	CQ--A-PE682-3
3313-	3314-	3341-	3342-	3369-	3370-	3397-	3398-	CQ--A-PE103-3
3315-	3316-	3343-	3344-	3371-	3372-	3399-	3400-	CQ--A-PE153-3
3317-	3318-	3345-	3346-	3373-	3374-	3401-	3402-	CQ--A-PE223-3
3319-	3320-	3347-	3348-	3375-	3376-	3403-	3404-	CQ--A-PE333-3
3321-	3322-	3349-	3350-	3377-	3378-	3405-	3406-	CQ--A-PE473-3
3323-	3324-	3351-	3352-	3379-	3380-	3407-	3408-	CQ--A-PE683-3
3325-	3326-	3353-	3354-	3381-	3382-	3409-	3410-	CQ--A-PE104-3
3327-	3328-	3355-	3356-	3383-	3384-	3411-	3412-	CQ--A-PE154-3
3501-	3502-	3531-	3532-	3561-	3562-	3591-	3592-	CQ--A-PF102-3
3503-	3504-	3533-	3534-	3563-	3564-	3593-	3594-	CQ--A-PF152-3
3505-	3506-	3535-	3536-	3565-	3566-	3595-	3596-	CQ--A-PF222-3
3507-	3508-	3537-	3538-	3567-	3568-	3597-	3598-	CQ--A-PF332-3
3509-	3510-	3539-	3540-	3569-	3570-	3599-	3600-	CQ--A-PF472-3
3511-	3512-	3541-	3542-	3571-	3572-	3601-	3602-	CQ--A-PF682-3
3513-	3514-	3543-	3544-	3573-	3574-	3603-	3604-	CQ--A-PF103-3
3515-	3516-	3545-	3546-	3575-	3576-	3605-	3606-	CQ--A-PF153-3
3517-	3518-	3547-	3548-	3577-	3578-	3607-	3608-	CQ--A-PF223-3
3519-	3520-	3549-	3550-	3579-	3580-	3609-	3610-	CQ--A-PF333-3
3521-	3522-	3551-	3552-	3581-	3582-	3611-	3612-	CQ--A-PF473-3
3523-	3524-	3553-	3554-	3583-	3584-	3613-	3614-	CQ--A-PF683-3
3525-	3526-	3555-	3556-	3585-	3586-	3615-	3616-	CQ--A-PF104-3
3527-	3528-	3557-	3558-	3587-	3588-	3617-	3618-	CQ--A-PF154-3
3529-	3530-	3559-	3560-	3589-	3590-	3619-	3620-	CQ--A-PF224-3

1/ In the MIL-C-19978/11 characteristic K superseding part number, there is no replacement for the .0010 to .15 in the 1000 volt dc range. In the MIL-C-19978/15 characteristic M superseding part number, there is no replacement for the .0010 to .068 in the 1000 volt dc range.

2/ Complete dash number will include, as the last digit, the applicable symbol to indicate the capacitance tolerance in percent.

3/ Complete part number will include the applicable symbol to represent style, circuit and applicable tolerance of the superseded dash number.

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TABLE V-A. Substitutability data.

Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ (D)
Dash numbers								MIL-C-19978/1
Style CQ08		Style CQ09		Style CQ12		Style CQ13		MIL-C-19978/1B
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	MIL-C-19978/1C MIL-C-19978/1D
3701-	3702-	3735-	3736-	3769-	3770-	3803-	3804-	CQ--A-TC152-3
3703-	3704-	3737-	3738-	3771-	3772-	3805-	3806-	CQ--A-TC392-3
3705-	3706-	3739-	3740-	3773-	3774-	3807-	3808-	CQ--A-TC472-3
3707-	3708-	3741-	3742-	3775-	3776-	3809-	3810-	CQ--A-TC822-3
3709-	3810-	3743-	3744-	3777-	3778-	3811-	3812-	CQ--A-TC103-3
3711-	3712-	3745-	3746-	3779-	3780-	3813-	3814-	CQ--A-TC123-3
3713-	3714-	3747-	3748-	3781-	3782-	3815-	3816-	CQ--A-TC153-3
3715-	3716-	3749-	3750-	3783-	3784-	3817-	3818-	CQ--A-TC183-3
3717-	3718-	3751-	3752-	3785-	3786-	3819-	3820-	CQ--A-TC223-3
3719-	3720-	3753-	3754-	3787-	3788-	3821-	3822-	CQ--A-TC273 3
3721-	3722-	3755-	3756-	3789-	3790-	3823-	3824-	CQ--A-TC333-3
3723-	3724-	3757-	3758-	3791-	3792-	3825-	3826-	CQ--A-TC393-3
3725-	3726-	3759-	3760-	3793-	3794-	3827-	3828-	CQ--A-TC473-3
3727-	3728-	3761-	3762-	3795-	3796-	3829-	3830-	CQ--A-TC563-3
3729-	3730-	3763-	3764-	3797-	3798-	3831-	3832-	CQ--A-TC683-3
3731-	3732-	3765-	3766-	3799-	3800-	3833-	3834-	CQ--A-TC823-3
3733-	3734-	3767-	3768-	3801-	3802-	3835-	3836-	CQ--A-TC104-3
3901-	3902-	3915-	3916-	3929-	3930-	3943-	3944-	CQ--A-TE104-3
3903-	3904-	3917-	3918-	3931-	3932-	3945-	3946-	CQ--A-TE124-3
3905-	3906-	3919-	3920-	3933-	3934-	3947-	3948-	CQ--A-TE154-3
3907-	3908-	3921-	3922-	3935-	3936-	3949-	3950-	CQ--A-TE184-3
3909-	3910-	3923-	3924-	3937-	3938-	3951-	3952-	CQ--A-TE224-3
3911-	3912-	3925-	3926-	3939-	3940-	3953-	3954-	CQ--A-TE274-3
3913-	3914-	3927-	3928-	3941-	3942-	3955-	3956-	CQ--A-TE334-3
4001-	4002-	4055-	4056-	4109-	4110-	4163-	4164-	CQ--A-TF102-3
4003-	4004-	4057-	4058-	4111-	4112-	4165-	4166-	CQ--A-TF122-3
4005-	4006-	4059-	4060-	4113-	4114-	4167-	4168-	CQ--A-TF152-3
4007-	4008-	4061-	4062-	4115-	4116-	4169-	4170-	CQ--A-TF182-3
4009-	4010-	4063-	4064-	4117-	4118-	4171-	4172-	CQ--A-TF222-3
4011-	4012-	4065-	4066-	4119-	4120-	4173-	4174-	CQ--A-TF272-3
4013-	4014-	4067-	4068-	4121-	4122-	4175-	4176-	CQ--A-TF332-3
4015-	4016-	4069-	4070-	4123-	4124-	4177-	4178-	CQ--A-TF392-3
4017-	4018-	4071-	4072-	4125-	4126-	4179-	4180-	CQ--A-TF472-3
4019-	4020-	4073-	4074-	4127-	4128-	4181-	4182-	CQ--A-TF562-3
4021-	4022-	4075-	4076-	4129-	4130-	4183-	4184-	CQ--A-TF682-3
4023-	4024-	4077-	4078-	4131-	4132-	4185-	4186-	CQ--A-TF822-3
4025-	4026-	4079-	4080-	4133-	4134-	4187-	4188-	CQ--A-TF103-3
4027-	4028-	4081-	4082-	4135-	4136-	4189-	4190-	CQ--A-TF123-3
4029-	4030-	4083-	4084-	4137-	4138-	4191-	4192-	CQ--A-TF153-3

See footnotes at end of table.

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TABLE V-A. Substitutability data - Continued.

Superseded dash number of MIL-C-19978/1A 1/								Superseded part number 2/
Dash numbers 3/								Style CQ08
Style CQ04		Style CQ05		Style CQ10		Style CQ11		Style CQ09 3/
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Style CQ12
								Style CQ13
4031-	4032-	4085-	4086-	4139-	4140-	4193-	4194-	CQ--A-TF183-3
4033-	4034-	4087-	4088-	4141-	4142-	4195-	4196-	CQ--A-TF223-3
4035-	4036-	4089-	4090-	4143-	4144-	4197-	4198-	CQ--A-TF273-3
4037-	4038-	4091-	4092-	4145-	4146-	4199-	4200-	CQ--A-TF333-3
4039-	4040-	4093-	4094-	4147-	4148-	4201-	4202-	CQ--A-TF393-3
4041-	4042-	4095-	4096-	4149-	4150-	4203-	4204-	CQ--A-TF473-3
4043-	4044-	4097-	4098-	4151-	4152-	4205-	4206-	CQ--A-TF563-3
4045-	4046-	4099-	4100-	4153-	4154-	4207-	4208-	CQ--A-TF683-3
4047-	4048-	4101-	4102-	4155-	4156-	4209-	4210-	CQ--A-TF823-3
4049-	4050-	4103-	4104-	4157-	4158-	4211-	4212-	CQ--A-TF104-3
4051-	4052-	4105-	4106-	4159-	4160-	4213-	4214-	CQ--A-TF124-3
4053-	4054-	4107-	4108-	4161-	4162-	4215-	4216-	CQ--A-TF154-3

1/ Complete dash number will include, as the last digit, the applicable symbol to indicate the capacitance tolerance in percent.
2/ Superseding part number will include the applicable symbol to represent style, circuit and applicable tolerance of the superseded dash number.
3/ Superseded styles CQ04, CQ05, CQ10 and CQ11 are replaced by styles CQ08, CQ09, CQ12 and CQ13 respectively.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR
Navy - AS, TD
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH

Preparing activity:
Air Force - 85

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